



Via MiEnviro Portal

July 6, 2026

Mr. Michael Cox
Environmental Quality Analyst – EGLE Air Quality Division
EGLE – Kalamazoo District Office
6938 Elm Valley Drive
Kalamazoo, MI 49009

RE: Response to June 15, 2026, Violation Notice Response
Pharmacia & Upjohn Company LLC, a subsidiary of Pfizer Inc. (Pfizer Kalamazoo)
Renewable Operating Permit Number ROP0000390v4.15

Dear Mr. Cox:

Pharmacia & Upjohn Company, LLC, a subsidiary of Pfizer Inc. (Pfizer Kalamazoo) received the above-referenced Violation Notice (VN) issued by Michigan's Department of Environment, Great Lakes, and Energy (EGLE) Air Quality Division (AQD) dated June 15, 2026 concerning EUC41NEOSPRAYDRYER-S3 T-26 W-RotoClone, a brand name for a wet particulate matter (PM) control device. The VN states that EGLE received Pfizer Kalamazoo's stack test report on May 4, 2026, and identified an exceedance of the PM emission limits in EUC41NEOSPRAYDRYER-S3 Special Condition (SC) I.1 and I.3 in the Renewable Operating Permit (ROP).

Background

On May 1, 2026, Pfizer Kalamazoo verbally notified EGLE's AQD that Pfizer Kalamazoo's stack test results indicated a potential exceedance of the ROP emission limits of EUC41NEOSPRAYDRYER-S3 T-26 W-RotoClone ("Stack T-26"). The stack test results were timely submitted to EGLE AQD Technical Programs Unit via MiEnviro Portal on May 4, 2026, and Pfizer Kalamazoo will report these results as a deviation in the upcoming Air ROP Semi-Annual Compliance Certification due to EGLE on September 15, 2026.

Pfizer takes its compliance obligations very seriously, and upon learning of the potential exceedance, Pfizer Kalamazoo calculated the potential fenceline emissions based on the stack-test results and confirmed that there were no adverse fenceline impacts.

NOV Response

Pursuant to the VN, the information requested by EGLE is provided below following restatements of EGLE's requests in bold text.

1. **Dates the violation occurred.** The stack test took place on March 5, 2026.
2. **An explanation of the causes.** As explained in Pfizer Kalamazoo's letter dated May 4, 2026 (attached again for your reference), Pfizer Kalamazoo promptly undertook several actions to determine why the test results deviated from the ROP emission limits. To date, Pfizer Kalamazoo and AAF International (AAF), the W-RotoClone manufacturer, have not identified a definite root cause.



3. **Duration of the violation.** The spray dryer operated on March 5, 2026, from 04:20 to 21:15, which is approximately 16.9 hours, during which time the stack test was conducted. The spray dryer also operated on March 9, 2026, from 03:00 to 20:55, which is approximately 17.9 hours. This operation occurred after the stack testing but prior to Pfizer Kalamazoo obtaining Stack T-26 stack testing results. There is no sampling data from this second operation, so it is unclear whether there was a deviation from the ROP emission limits.
4. **Whether the violation is ongoing.** The violation is not ongoing. The Neomycin Spray Dryer process has not operated since March 9, 2026, at 20:55. The process will not restart until the issue is resolved.
5. **A summary of the actions that have been taken and are proposed to be taken to correct the violation and the dates by which these actions will take place.** Pfizer Kalamazoo undertook a thorough investigation to determine why the test results deviated from the ROP emission limits. As described above, AAF and Pfizer Kalamazoo worked collaboratively to review the functional operation of the Stack T-26 W-RotoClone. During these inspections, only air and water were run through the process. To date, AAF and Pfizer Kalamazoo have not determined why the W-RotoClone may not be operating at the guaranteed efficiency based on the available data and status of the equipment.

Pfizer Kalamazoo also developed corrective and preventative actions (CAPAs) to correct the deviation. One of these CAPAs included submission of a permit-to-install (PTI) application on May 4, 2026, to request an increase in the hourly and pound per thousand pounds of exhaust air limits without increasing the allowed pounds per month. The PTI application demonstrates that there are no adverse impacts at the fence line as discussed above and there will be no net increase in PM emissions from the site. The rest of the CAPAs are described below.

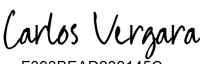
6. **Steps that are being taken to prevent a recurrence.** Pfizer Kalamazoo developed several actions to prevent a recurrence. Each of the CAPAs were added to Pfizer's CAPA tracking system to track closure and must be completed prior to the unit returning to operation. In addition to the submission of the PTI application described above, these CAPAs include:
 - The T-26 W-RotoClone water cone-to-impeller distance was adjusted to within the manufacturer's specification.
 - The annual preventative maintenance workflow is being updated for the T-26 W-RotoClone to include verification that the water cone-to-impeller distance is within the manufacturer's specification.
 - The manufacturer only recommends confirming this measurement for certain maintenance activities, but it is being added as a double check that the impeller and water cone have not shifted during normal operation.



7000 Portage Road
Kalamazoo, MI 49001-0199

Pfizer remains committed to leading in environmental stewardship. Consistent with that commitment, Pfizer conducted a thorough investigation of this matter, including an evaluation of potential causes, environmental impacts, and opportunities for corrective and preventive action. If you need additional information or have any questions, please feel free to contact Timothy Swainston at 269.833.0080 or timothy.swainston@pfizer.com and/or Jeffrey Robey at 269.833.3842 or jeffrey.robey@pfizer.com.

Sincerely,

DocuSigned by:

F393BEAD239145C...

Carlos Vergara
Site Leader

Pfizer Global Supply
Kalamazoo Site Leader
O: 269.833.9805
carlos.e.vergara@pfizer.com

Enclosure



May 4, 2026

Ms. Monica Brothers
Michigan Department of Environmental Quality
Air Quality Division – Kalamazoo District Office
6938 Elm Valley Drive
Kalamazoo, MI 49009

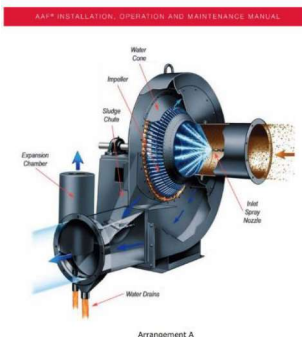
RE: Neomycin Spray Dryer T-26 W-RotoClone Particulate Matter Emissions Compliance Test Results

Dear Ms. Brothers:

On behalf of Pharmacia and Upjohn Company LLC, a subsidiary of Pfizer Inc. ("Pfizer Kalamazoo"), please find enclosed the results from the March 5, 2026 particulate matter ("PM") emissions compliance test performed by TRC Environmental Corporation ("TRC") on the Neomycin Spray Dryer ("Stack T-26") wet dust collector pursuant to Special Condition V.1. The emissions compliance test report identified a result that deviates from the Renewable Operating Permit ("ROP") ROP0000390v4.14's pound per hour and pound per thousand pounds of exhaust air limits for PM. Pfizer Kalamazoo will report these results as deviations in the upcoming Air ROP Semi-Annual Compliance Certification due September 15, 2026.

Upon receiving the draft test results, Pfizer Kalamazoo confirmed that the Neomycin Sulfate process using the Neomycin Spray Dryer would not run again until these findings were resolved. Pfizer Kalamazoo promptly undertook the following actions to determine why the test results deviated from the ROP emission limits and confirm there was no adverse impact to human health or the environment:

- Manufacturer's Review: AAF International ("AAF"), the manufacturer of the W-RotoClone, physically inspected the T-26 W-RotoClone on April 14, 2026.
 - The final inspection report is pending and AAF verbally reported that the unit is in good condition. They have not yet been able to determine the cause of the deviation from the specified exhaust limit.
 - AAF noted that the distance between the impeller and the water cone was out of specification. Please see image below. AAF has not yet determined if this would impact control efficiency.
- The shift of the impeller clearance distance could have occurred during the replacement of the fan belts, performed as part of Pfizer Kalamazoo's routine preventative maintenance on February 23, 2026.
- Because AAF does not address this specification in its operations manual for routine service checks of the equipment, Pfizer Kalamazoo does not routinely check the impeller clearance distance during its preventative maintenance.
- The last annual preventative maintenance on the W-RotoClone was completed by Pfizer Kalamazoo on February 23rd.
- The distance of the impeller to the water cone will be adjusted to the manufacturer's specifications prior to resuming operations. Pfizer Kalamazoo will add this check to its preventative maintenance procedures whether or not it was the cause of deviation from the required control efficiency.

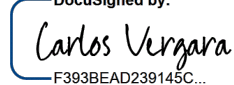




- Confirmation of Inlet Loading: Pfizer Kalamazoo verified that the maximum inlet loading to the W-RotoClone was within the manufacturer's specification via mass balance. Using the yield data from the tested lot, Pfizer Kalamazoo estimated that the inlet loading during testing was approximately 7.2 pounds per hour versus the manufacturer's allowed maximum loading to the W-RotoClone of 124.1 pounds per hour. Therefore, the actual loading was estimated to 5.8% of the manufacturer's allowed maximum loading.
- Particle Size Distribution Testing: Pfizer Kalamazoo completed particulate size distribution testing of both the Neomycin Sulfate product packaged and the Neomycin Sulfate collected during the stack testing by TRC on March 5, 2026, to determine the particulate size in microns (particulate size length). Based on the particle size testing results and the AAF published collection efficiency table, the measured emissions from Stack T-26 should have met the ROP emission limits.
- Compliance with Monthly Allowable Emission Limit: Using the maximum one-hour emission test results from Stack T-26, Pfizer Kalamazoo calculated the estimated monthly emissions from Stack T-26 in January and February 2026. These calculations confirm that Pfizer Kalamazoo would not have exceeded the ROP's 740 pound per month emission limit if the test results reflect operations during those months, and in fact only emitted approximately 33% of the monthly allowable emission limit.
- No Adverse Impacts at the Fenceline: Pfizer Kalamazoo also calculated the potential fenceline emissions based on the stack test results, and confirmed that there are no adverse fenceline impacts, assuming an annual screening level of 27 micrograms per cubic meter based on the calculated ITSL.
- Corrective Action: We will continue to work with AAF to evaluate the effectiveness of our control device. As a precautionary measure, Pfizer Kalamazoo is submitting a Permit-to-Install application alongside this submittal to request an increase in the hourly and pound per thousand pounds of exhaust air limits without increasing the allowed pounds per month. This PTI application demonstrates that there are no adverse impacts at the fenceline as discussed above.

Pfizer Kalamazoo remains committed to leading in environmental stewardship and to full compliance with all regulatory requirements. We will continue to work with AAF and EGLE's Air Quality Division to resolve this issue. If you need additional information or have any questions, please feel free to contact Timothy Swainston at 269.833.0080 or timothy.swainston@pfizer.com.

Sincerely,

DocuSigned by:

F393BEAD239145C...

Carlos Vergara
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Enclosure